

Exp 33-Scatter Plot with Empty Circles Using Random X and Y Values

Aim

To write a Python program using Matplotlib to draw a scatter plot with empty circles using randomly generated X and Y values.

Procedure

1. Import numpy and matplotlib.pyplot.
2. Generate random values for X and Y.
3. Use plt.scatter() to create the scatter plot.
4. Set facecolors='none' to make the circles empty.
5. Set edgecolors='red' to show the circle boundaries.
6. Add X-axis, Y-axis labels and title.
7. Display the graph.

Code

```
33.py - C:/Users/subha/OneDrive/Desktop/Query processing/33.py (3.12.10)
File Edit Format Run Options Window Help
import numpy as np
import matplotlib.pyplot as plt

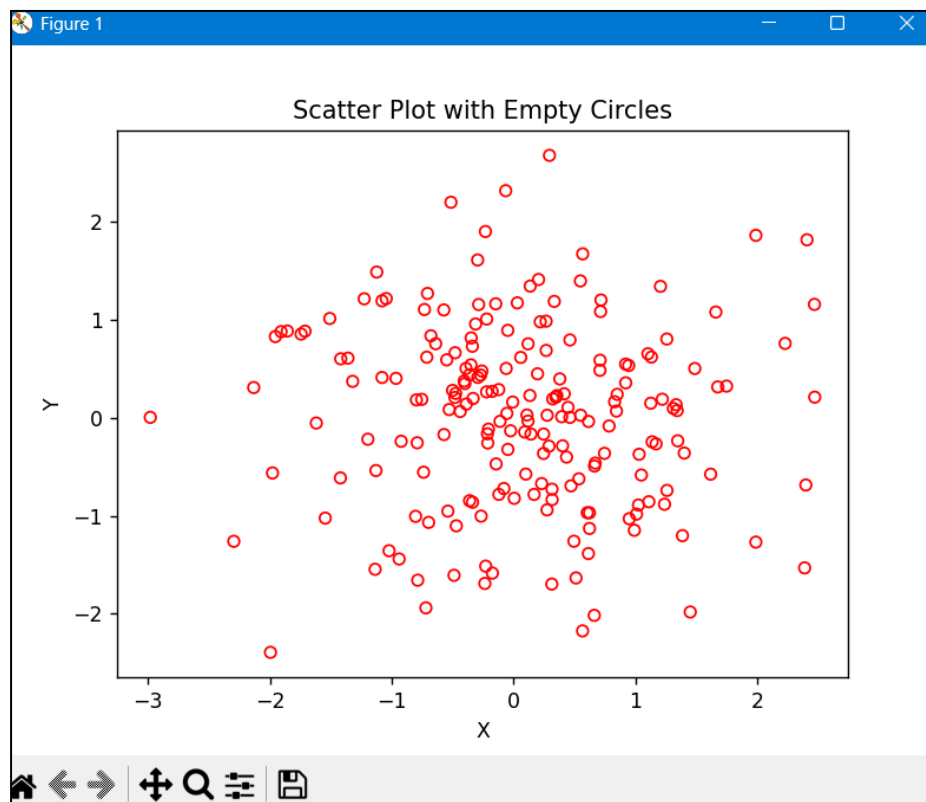
# Generate random data
np.random.seed(10)
x = np.random.randn(200)
y = np.random.randn(200)

# Scatter plot with empty circles
plt.scatter(x, y, facecolors='none', edgecolors='red', s=30)

# Labels and title
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Empty Circles')

# Display plot
plt.show()
```

Output



Result

Thus, the Python program successfully creates a scatter plot with empty circles using random X and Y distributions.

Exp 34-Scatter Plot with Balls of Different Sizes

Aim

To write a Python program using Matplotlib to create a scatter plot with randomly generated balls of different sizes and colors.

Procedure

1. Import numpy and matplotlib.pyplot.
2. Generate random X and Y coordinates.
3. Generate random sizes for the balls.
4. Generate random colors for the balls.
5. Use `plt.scatter()` with the `s` parameter to set different ball sizes.
6. Add X-axis, Y-axis labels and title.
7. Display the scatter plot.

Code

```
34.py - C:/Users/subha/OneDrive/Desktop/Query processing/34.py (3.12.10)
File Edit Format Run Options Window Help
import numpy as np
import matplotlib.pyplot as plt

# Generate random data
np.random.seed(10)

x = np.random.rand(25)
y = np.random.rand(25)

# Generate different sizes
sizes = np.random.randint(100, 1000, 25)

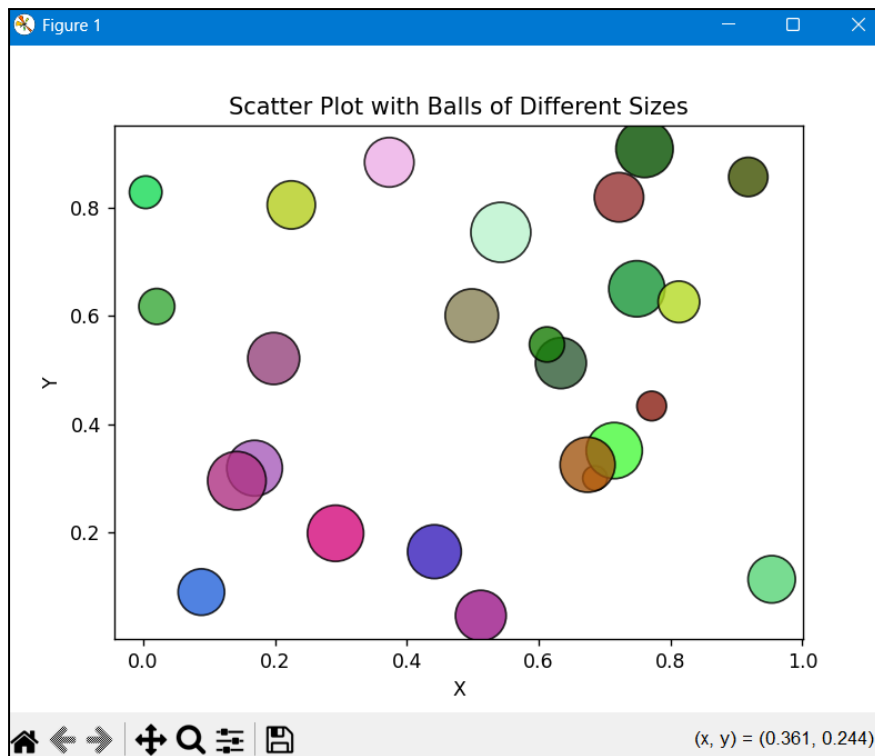
# Generate random colors
colors = np.random.rand(25, 3)

# Create scatter plot
plt.scatter(x, y, s=sizes, c=colors, alpha=0.8,
            edgecolors='black')

# Labels and title
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Balls of Different Sizes')

# Display plot
plt.show()
```

Output



Result

Thus, the Python program successfully creates a scatter plot with balls of different sizes using random distributions.

Exp 35-Comparison of Mathematics and Science Marks Using Scatter Plot

Aim

To write a Python program using Matplotlib to draw a scatter plot comparing the Mathematics and Science marks of 10 students.

Procedure

1. Import the matplotlib.pyplot library.
2. Store Mathematics and Science marks of 10 students.
3. Define the marks range for the X-axis.
4. Plot Mathematics marks using `plt.scatter()`.
5. Plot Science marks using `plt.scatter()`.
6. Use different colors to distinguish the two subjects.
7. Add labels, title, and legend.
8. Display the scatter plot using `plt.show()`.

Code

```
35.py - C:/Users/subha/OneDrive/Desktop/Query processing/35.py (3.12.10)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

# Sample data
math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

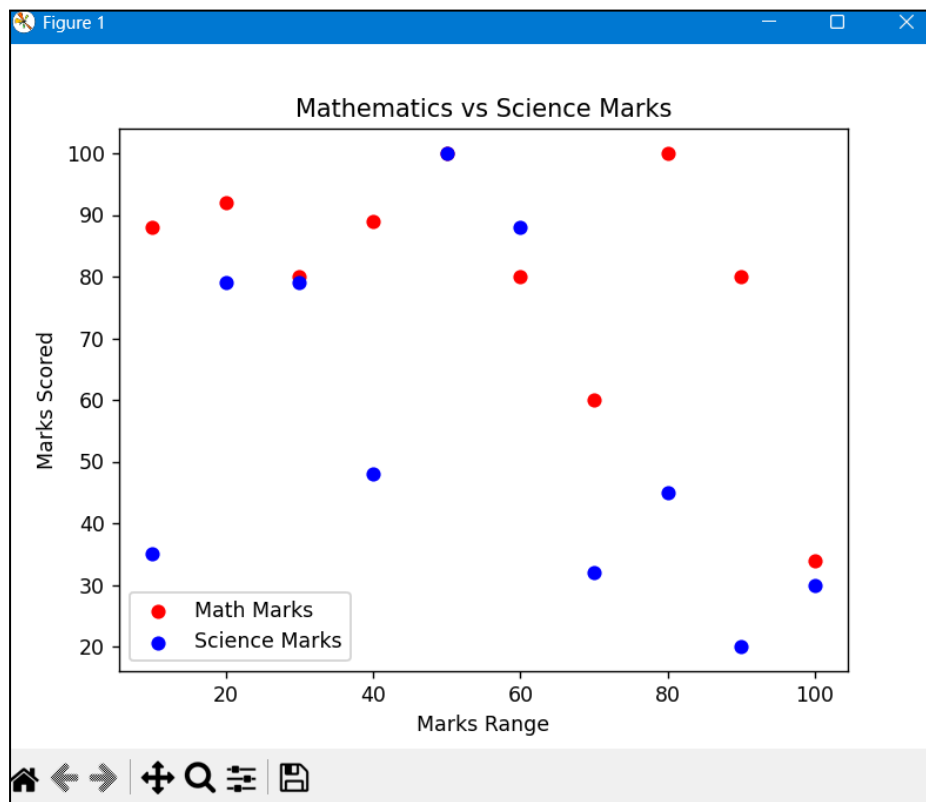
# Scatter plots
plt.scatter(marks_range, math_marks, color='red', label='Math Marks')
plt.scatter(marks_range, science_marks, color='blue', label='Science Marks')

# Labels and title
plt.xlabel('Marks Range')
plt.ylabel('Marks Scored')
plt.title('Mathematics vs Science Marks')

# Legend
plt.legend()

# Display graph
plt.show()
```

Output



Result

Thus, the Python program successfully creates a scatter plot comparing Mathematics and Science marks of 10 students.